

Exp No: 10

PIR SENSOR IN PROTEUS USING ARDUINO

OBJECTIVES

- To interface a PIR (Passive Infrared) sensor with Arduino UNO using Proteus simulation.
- To detect human motion using the PIR sensor.
- To control an LED and display the PIR sensor status through the Arduino and Virtual Terminal.

PROGRAM

```
int Status = 13; // Digital pin D6

int sensor = 12; // Digital pin D7

void setup() {
  Serial.begin(9600);
  pinMode(sensor, INPUT); // declare sensor as input
  pinMode(Status, OUTPUT); // declare LED as output
}

void loop() {
  long state = digitalRead(sensor);
  Serial.println(state);
  if(state == HIGH) {
    digitalWrite (Status, HIGH);
    Serial.println("Motion detected!");
    delay(1000);
  }
  else {
    digitalWrite (Status, LOW);
    Serial.println("Motion absent!");
    delay(1000);
  }
}
```

}

OUTPUT

